



Lesson 4.2 — Vertical Asymptotes and Holes

Big idea: Vertical asymptotes are walls where the function blows up to $\pm\infty$. Holes are single missing points. The difference is whether the troublesome factor cancels or not.

Quick review

Step 1: factor numerator and denominator completely.

Step 2: any common factor $\rightarrow$ produces a HOLE at the x that makes that factor zero.

Step 3: any factor that remains only in the denominator $\rightarrow$ produces a VERTICAL ASYMPTOTE.

Hole height: after cancelling the common factor, plug the x back into the simplified expression to find the y -coordinate of the hole.

Worked example

Worked example. Identify holes and VAs of $f(x) = (x^2 - 4)/[(x - 2)(x^2 - x - 6)]$.

Factor: $(x - 2)(x + 2) / [(x - 2)(x - 3)(x + 2)]$. Common factors $(x - 2)$ and $(x + 2)$ cancel.

Holes: $x = 2$ and $x = -2$. VA: $x = 3$. Simplified: $1/(x - 3)$. Hole heights: at $x = 2 \rightarrow 1/(-1) = -1$; at $x = -2 \rightarrow 1/(-5) = -1/5$.

Warm-ups

1. Find the vertical asymptote(s) of $f(x) = 3/(x - 7)$.
2. Find the hole of $f(x) = (x - 1)(x + 2)/(x + 2)$.
3. Find the vertical asymptotes of $f(x) = (x + 1)/(x^2 - 4)$.

Practice

1. For $f(x) = (x^2 - x - 6)/(x^2 - 9)$, find any holes and VAs and the simplified form.
2. For $f(x) = (x^3 - x)/[x(x - 1)^2]$, find holes and VAs and give the y -coordinate of any hole.
3. Construct a rational function with a hole at $(4, -2)$ and a vertical asymptote at $x = -1$.

Challenge

1. Graph (describe) the local behavior of $f(x) = (x - 3)/(x^2 - 9)$ at its hole and VA.
2. Why can a rational function have at most as many vertical asymptotes as the degree of its denominator (after cancellation)?



Answer key — Lesson 4.2

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $x = 7$.
2. Hole at $x = -2$.
3. $x^2 - 4 = (x - 2)(x + 2)$. VAs at $x = 2$ and $x = -2$.

Practice

1. Factor: $(x - 3)(x + 2) / [(x - 3)(x + 3)]$. $(x - 3)$ cancels $\rightarrow$ hole at $x = 3$. VA at $x = -3$. Simplified: $(x + 2)/(x + 3)$. Hole height: $5/6$.
2. Factor: $x(x - 1)(x + 1) / [x(x - 1)^2]$. x and one $(x - 1)$ cancel. Holes: $x = 0$ (height = $(0 + 1)/(0 - 1)$ using simplified $(x + 1)/(x - 1) \rightarrow -1$) and $x = 1$ if $(x - 1)$ cancelled completely (it didn't — denominator still has $(x - 1)^1$ left). So at $x = 1$, VA, not hole. Result: hole at $x = 0$ (height -1), VA at $x = 1$.
3. $f(x) = (x - 4) \cdot (x - a) / [(x + 1) \cdot (x - 4)]$ for some a , then evaluate at $x = 4$ in simplified form $(x - a)/(x + 1)$ and set it equal to -2 : $(4 - a)/5 = -2 \Rightarrow a = 14$. So $f(x) = (x - 4)(x - 14) / [(x + 1)(x - 4)]$.

Challenge

1. Factor: $(x - 3)/[(x - 3)(x + 3)] = 1/(x + 3)$. Hole at $x = 3$, height $1/6$. VA at $x = -3$ ($1/(x+3)$ blows up).
2. Vertical asymptotes occur at zeros of the simplified denominator. A polynomial of degree d has at most d real zeros, so the denominator (after cancellation) contributes at most d VAs.